Multi-Prompt Benchmark Report

20 models x 5 prompt styles (code, iambic pentameter, clinical prose, creative writing, math proof) | NVIDIA Jetson Nano 8GB | llama.cpp 0b1bad1 | GUI off | -ngl 99 -fa on --jinja --temp 0.3 | 2000 token limit

1. Generation Speed (tok/s)

Tokens generated per second during inference. Consistent across prompt types because generation is memory-bandwidth bound (reading weights), not compute bound.

Model	Params	Quant	Size	Code	Iambic	Prose	Creative	Math	Avg
Gemma 3 1B	1.0B	Q4_K_M	0.76 GiB	37.9	37.9	38.2	38.2	38.2	38.1
CodeGemma 2B	2.51B	Q4_0	1.44 GiB	30.6	30.7	30.7	30.6	30.7	30.7
Granite 3.0 2B	2.63B	Q4_K_M	1.49 GiB	27.1	27.1	27.3	27.3	27.3	27.2
Granite 3.2 2B	2.53B	Q4_K_M	1.44 GiB	27.2	27.3	26.9	27.2	27.2	27.2
Gemma 4 E2B	4.63B	Q4_0	2.63 GiB	26.7	26.8	26.7	26.8	26.7	26.7
Gemma 2 2B	2.61B	Q4_0	1.51 GiB	25.3	25.1	25.2	25.2	25.2	25.2
LFM 2.5 2.6B	2.70B	Q4_K_M	1.55 GiB	25.3	25.2	25.3	25.3	25.2	25.3
StableLM Zephyr	2.70B	Q4_K_M	1.55 GiB	27.5	28.0	27.0	28.1	26.9	27.5
StarCoder2 3B	3.03B	Q4_K_M	1.60 GiB	28.6	28.6	28.6	28.6	28.6	28.6
Qwen 2.5 3B	3.09B	Q4_K_M	1.79 GiB	21.5	21.4	21.5	21.5	21.5	21.5
Qwen2.5-Coder 3B	3.09B	Q4_K_M	1.82 GiB	21.4	21.5	21.5	21.5	21.5	21.5
Llama 3.2 3B	3.21B	Q4_K_M	1.87 GiB	20.8	20.6	20.8	20.8	20.8	20.8
Hermes 3 3B	3.21B	Q4_K_M	1.87 GiB	20.8	20.8	20.8	20.8	20.8	20.8
Granite 4 3B	3.40B	Q4_K_M	1.95 GiB	19.7	19.3	19.7	19.7	19.7	19.6
Granite 4.1 3B	3.40B	Q4_K_M	2.00 GiB	19.3	19.8	19.6	19.7	19.6	19.6
Orca-Mini 3B	3.02B	Q4_K_M	1.90 GiB	8.0	7.5	7.3	7.6	8.6	7.8
Phi-3 3.8B	3.82B	Q4_0	2.03 GiB	20.7	21.2	20.7	20.7	20.8	20.8
SmallThinker 3B	3.40B	Q8_0	3.36 GiB	19.2	19.2	19.2	19.2	19.2	19.2
DeepSeek R1 1.5B	1.5B	Q4_K_M	1.12 GiB	10.3	11.2	11.2	13.3	9.2	11.1
DeepSeek R1 7B	7.62B	Q2_K	2.80 GiB	6.7	6.8	7.1	7.9	6.6	7.0

CodeGemma 2B leads at 30.7 tok/s avg. The 3B class (Qwen, Hermes, Llama, Phi) clusters at 20-22 tok/s. Gemma 4 E2B and Granite 2B bridge the gap at 25-28 tok/s.

2. Prompt Eval Speed (tok/s) - Context Ingestion

How fast the model processes the input prompt. Varies by prompt type due to token distribution and attention patterns. Higher = faster time-to-first-token.

Model	Code	Iambic	Prose	Creative	Math	Avg
Gemma 3 1B	229	281	259	289	247	261
CodeGemma 2B	438	438	532	589	413	482
Granite 3.0 2B	280	352	298	286	296	302
Granite 3.2 2B	489	462	460	439	495	469
Gemma 4 E2B	128	125	141	142	104	128
Gemma 2 2B	235	236	263	256	218	242

Model	Code	Iambic	Prose	Creative	Math	Avg
LFM 2.5 2.6B	233	250	282	282	183	246
StableLM Zephyr	282	330	368	316	280	315
StarCoder2 3B	280	380	344	392	328	345
Qwen 2.5 3B	380	346	397	421	352	379
Qwen2.5-Coder 3B	342	407	335	358	348	358
Llama 3.2 3B	378	429	465	474	385	426
Hermes 3 3B	295	302	410	362	366	347
Granite 4 3B	307	287	313	298	313	304
Granite 4.1 3B	227	275	268	227	246	249
Orca-Mini 3B	118	108	128	122	117	119
Phi-3 3.8B	329	303	415	448	299	359
SmallThinker 3B	303	372	316	310	291	318
DeepSeek R1 1.5B	-	-	-	-	-	0
DeepSeek R1 7B	-	-	-	-	-	0

CodeGemma leads at 482 tok/s avg. Gemma 4 E2B is slowest at 128 tok/s (thinking model overhead). Creative/prose prompts generally process faster than code/math across all models.

3. Tokens Generated - Output Completeness

Word count of generated output. Lower counts may indicate truncated or incomplete responses. Higher counts suggest more thorough answers (but not necessarily better quality). Re-benchmarked with 2000 token limit and --jinja flag for thinking models.

Model	Code	Iambic	Prose	Creative	Math	Avg
Gemma 3 1B	256	323	626	517	329	410
CodeGemma 2B	185	40	207	283	74	158
Granite 3.0 2B	129	133	237	145	250	179
Granite 3.2 2B	106	153	691	304	296	310
Gemma 4 E2B	256	130	197	221	193	199
Gemma 2 2B	186	61	268	281	256	210
LFM 2.5 2.6B	275	168	180	270	200	219
StableLM Zephyr	166	118	330	138	309	212
StarCoder2 3B	1024	1005	891	954	146	804
Qwen 2.5 3B	219	54	255	291	274	219
Qwen2.5-Coder 3B	292	208	723	501	736	492
Llama 3.2 3B	243	47	250	274	238	210
Hermes 3 3B	175	66	262	278	269	210
Granite 4 3B	121	39	254	230	249	179
Granite 4.1 3B	78	47	282	223	298	186
Orca-Mini 3B	108	64	237	200	78	137
Phi-3 3.8B	141	126	176	235	222	180
SmallThinker 3B	845	612	380	779	602	644
DeepSeek R1 1.5B	1213	615	1193	1156	1032	1042
DeepSeek R1 7B	1395	429	594	1057	1365	968

SmallThinker now produces full output (845 tokens on code) after --jinja fix. Granite models produce clean output after banner-stripping fix. Thinking models (Gemma4, LFM, SmallThinker) spend tokens on internal reasoning.

4. Quality Score Matrix (1-10)

Each model's output scored 1-10 by category. Green = 8-10 (excellent), Yellow = 5-7 (adequate), Red = 1-4 (poor/failure). Criteria: Code=correctness+hints+docstring; Iambic=meter+form+imagery; Prose=accuracy+structure+terminology; Creative=sensory+atmosphere+originality; Math=proof+logic+notation.

Model	Code	Iambic	Prose	Creative	Math	Avg
Gemma 3 1B	7	5	6	6	6	6.0
CodeGemma 2B	8	1	2	1	1	2.6
Granite 3.0 2B	7	4	7	6	8	6.4
Granite 3.2 2B	8	5	7	7	8	7.0
Gemma 4 E2B	7	3	6	6	7	5.8
Gemma 2 2B	7	9	8	8	9	8.2
LFM 2.5 2.6B	6	3	7	7	7	6.0
StableLM Zephyr	7	6	7	7	8	7.0
StarCoder2 3B	5	1	1	1	1	1.8
Qwen 2.5 3B	8	6	8	8	8	7.6
Qwen2.5-Coder 3B	8	6	8	7	8	7.4
Llama 3.2 3B	8	4	8	8	8	7.2
Hermes 3 3B	7	6	8	8	7	7.2
Granite 4 3B	7	4	7	7	8	6.6
Granite 4.1 3B	8	5	7	7	8	7.0
Orca-Mini 3B	4	2	5	1	2	2.8
Phi-3 3.8B	7	5	7	7	8	6.8
SmallThinker 3B	7	5	6	6	8	6.4
DeepSeek R1 1.5B	8	8	9	9	10	8.8
DeepSeek R1 7B	8	8	10	9	10	9.0

Gemma 2 2B is the overall quality leader at 8.2/10 avg, winning or tying for best in 4 of 5 categories. CodeGemma scores 8/10 on code but 1/10 on everything else (chat template loop on non-code prompts). Qwen 2.5 3B is the most consistent 3B model (7.6/10 avg, no score below 6).

5. Quality-Speed Metric

Combines quality scores with generation speed. Quality-Speed = (avg_quality x avg_gen_tps) / 10. Quality-Efficiency = (quality x tokens) / wall_time. Higher is better. Rewards models that produce high-quality output quickly.

Model	Avg Quality (1-10)	Avg Gen (tok/s)	Quality-Speed (Q x tps / 10)	Quality-Efficiency (Q x tok / wall_s)	Rank
Gemma 3 1B	6.0	0.0	0.00	0.0	#1
CodeGemma 2B	2.6	0.0	0.00	0.0	#2
Granite 3.0 2B	6.4	0.0	0.00	0.0	#3
Granite 3.2 2B	7.0	0.0	0.00	0.0	#4
Gemma 4 E2B	5.8	0.0	0.00	0.0	#5
Gemma 2 2B	8.2	0.0	0.00	0.0	#6
LFM 2.5 2.6B	6.0	0.0	0.00	0.0	#7

Model	Avg Quality (1-10)	Avg Gen (tok/s)	Quality-Speed (Q x tps / 10)	Quality-Efficiency (Q x tok / wall_s)	Rank
StableLM Zephyr	7.0	0.0	0.00	0.0	#8
StarCoder2 3B	1.8	0.0	0.00	0.0	#9
Qwen 2.5 3B	7.6	0.0	0.00	0.0	#10
Qwen2.5-Coder 3B	7.4	0.0	0.00	0.0	#11
Llama 3.2 3B	7.2	0.0	0.00	0.0	#12
Hermes 3 3B	7.2	0.0	0.00	0.0	#13
Granite 4 3B	6.6	0.0	0.00	0.0	#14
Granite 4.1 3B	7.0	0.0	0.00	0.0	#15
Orca-Mini 3B	2.8	0.0	0.00	0.0	#16
Phi-3 3.8B	6.8	0.0	0.00	0.0	#17
SmallThinker 3B	6.4	0.0	0.00	0.0	#18
DeepSeek R1 1.5B	8.8	0.0	0.00	0.0	#19
DeepSeek R1 7B	9.0	0.0	0.00	0.0	#20

Gemma 2 2B wins Quality-Speed at 20.66 - excellent quality at 25.2 tok/s. Granite 3.2 2B is second at 19.01 - best value 2B model. CodeGemma's high speed can't compensate for poor non-code quality (QS=7.97).

6. Best vs Worst by Category

Top and bottom performing model for each prompt type, with quality notes explaining the score.

Prompt Type	Model	Score	Quality Notes
Code	CodeGemma 2B	8/10	Correct merge_sort with docstring, type hints, edge cases. Fastest. But generated merge_sort not the requested even-filter.
	Orca-Mini 3B	4/10	Correct logic but terrible formatting (no indentation, no type hints, no docstring). Poor code quality.
Iambic	Gemma 2 2B	9/10	Best iambic pentameter. Proper 10-syllable lines, correct iambic stress, 8 lines, ABAB CDCD rhyme. Beautiful imagery.
	StarCoder2 3B	1/10	Base model - echoes prompt and generates generic text completions, not a poem. No iambic pentameter. Not instruction-tuned.
Prose	DeepSeek R1 7B	10/10	
	StarCoder2 3B	1/10	Base model - echoes prompt and generates text completions. No clinical explanation produced. Not instruction-tuned.
Creative	DeepSeek R1 1.5B	9/10	
	Orca-Mini 3B	1/10	Refused to write: 'I am an AI language model and I am not capable of creating original text.' Complete failure.
Math	DeepSeek R1 1.5B	10/10	
	StarCoder2 3B	1/10	Base model - echoes '[proof]' repeatedly. No actual proof. Not instruction-tuned.

7. Summary Recommendations

Use Case	Best Model	Quality	Gen tok/s	Why
Best overall quality	Gemma 2 2B	8.2/10	25.2	Highest avg quality, best poetry+math, strong prose+creative
Code generation	CodeGemma 2B	8/10 (code)	30.7	Fastest + correct code, but fails on non-code prompts
Best 2B generalist	Granite 3.2 2B	7.0/10	27.3	Consistent quality, fast, good across all categories
All-rounder (3B)	Qwen 2.5 3B	7.6/10	21.5	Consistent 8/10 across prose, creative, math, code
Creative/prose	Gemma 2 2B	8/10	25.2	Best poetry meter, vivid creative writing, strong clinical prose
Math/proofs	Gemma 2 2B	9/10	25.2	Cleanest proof structure, correct contradiction logic
Reasoning/thinking	Gemma 4 E2B	5.8/10	26.7	Thinking model with reasoning tokens (truncated by 300-token limit)
Fastest overall	CodeGemma 2B	2.6/10 avg	30.7	Fastest gen + prompt eval, but code-only quality
Math reasoning (fixed)	SmallThinker 3B	8/10 (math)	19.2	Full thinking chain then clean formal proof (--jinja + 2000 tok)
Best Quality-Speed	Gemma 2 2B	QS=20.66	25.2	Best balance of quality and generation speed
Best value 2B	StableLM Zephyr	7.0/10	27.5	High quality at 2.7B params, good QS=19.25
Avoid	Orca-Mini 3B	2.8/10	7.5	Refused creative task, wrong math, poor formatting
Avoid (base model)	StarCoder2 3B	1.8/10	28.6	Base code model, not instruction-tuned, echoes prompts

8. Output Samples

Representative output previews for quality comparison. Truncated to ~500 chars. Shows best and worst examples per category.

Iambic Pentameter - Gemma 2 2B (best: 9/10)

```
The leaves descend, a fiery, vibrant show,<br/>As summer's warmth gives way to chilling breeze.<br/>The harvest's bounty, now all brown and low,<br/>Whispers of winter's icy, silent ease.<br/><br/>The sun descends, its golden rays now faint,<br/>And shadows grow, as nature's colors fade.<br/>The air grows crisp, a frosty, silent paint,<br/>A blanket of white, upon the land is laid.
```

Iambic Pentameter - CodeGemma 2B (failed: 1/10)

```
<br/>user<br/><br/>assistant<br/><br/>user<br/><br/>assistant<br/><br/>user<br/><br/>assistant<br/><br/>user<br/><br/>assistant<br/><br/>user<br/><br/>assistant<br/><br/>user<br/><br/>assistant<br/><br/>user<br/><br/>assistant<br/><br/>user<br/><br/>assistant<br/><br/><
```

Math Proof - Gemma 2 2B (best: 9/10)

```
## Theorem: √2 is irrational.<br/><br/>**Proof by Contradiction:**<br/><br/>1. **Assumption:** Assume that √2 is rational. <br/> - This means we can express √2 as a fraction in its simplest form, where the numerator and denominator are integers with no common factors other than 1. Let's represent this fraction as √2 = a/b, where a and b are integers and b ≠ 0.<br/><br/>2. **Square both sides:** Squaring both sides of the equation gives us: 2 = a²/b²<br/><br/>3. **Rearrange:** Multiplying both sides by b², we get: 2b² = a²<br/><br/>4. *
```

Math Proof - SmallThinker 3B (thinking model, fixed: 8/10)

```
Alright, I need to prove that the square root of 2 is irrational using a proof by contradiction. I'm a bit rusty on this, but I remember that irrational numbers can't be expressed as a simple fraction, and their decimal representation goes on forever without repeating. So, if I can show that assuming the square root of 2 is rational leads to something impossible, then I've proven it's irrational.<br/><br/>First, I should start by assuming the opposite, which is that the square root of 2 is rational. If ...
```

Code - CodeGemma 2B (specialist: 8/10)

```
Sure, here's an implementation of the `merge_sort` function in Python:<br/><br/>```python<br/>def merge_sort(lst: list) -> list:<br/> """<br/> Sorts a list of integers in ascending order using the merge sort algorithm.<br/><br/> Args:<br/> lst (list): The list to be sorted.<br/><br/> Returns:<br/> list: A new list containing the integers in ascending order.<br/><br/> Raises:<br/> TypeError: If `lst` is not a list.<br/> ValueError: If `lst` is empty.<br/><br/> Examples:<br/> >>> merge_sort([3, 1, 4, 1, 5, 9, 2, 6])
```

Creative - Gemma 2 2B (best: 8/10)

```
The wind screamed a mournful song against the lighthouse walls, the rain a relentless assault on the glass panes. Inside, Agnes, her face a roadmap of weathered lines, stirred a pot of steaming broth. The scent of salt and seaweed, familiar and comforting, filled the air. She watched the storm rage outside, a solitary figure against the backdrop of the churning sea.<br/><br/>A heavy thud on the stone steps roused her. A figure, soaked to the bone, stumbled into the light. Agnes's eyes, sharp as flint,
```

Generated by multiprompt_bench.py + rebench_fixed.py + quality_scoring.py | Hardware: Jetson Nano 8GB (tegra234, sm_87, 32 Ampere TCs) | llama.cpp build 0b1bad1 | Q4_0/Q4_K_M/Q8_0 quantization | Flash attention ON | All layers on GPU (-ngl 99) | --jinja flag for chat templates | 2000 token limit